

February 23, 2026

The Honorable Thomas Keane, M.D.

Assistant Secretary for Technology Policy
Office of the Assistant Secretary for Technology Policy (ASTP)
Mary E. Switzer Building
Mail Stop: 7033A
330 C Street S.W.
Washington, D.C. 20201

RE: Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as Part of Clinical Care

Dear Dr. Keane:

The Healthcare Leadership Council (HLC) and the Confidentiality Coalition appreciate the opportunity to submit comments to the Office of the Deputy Secretary and Assistant Secretary for Technology Policy (ASTP) and Office of the National Coordinator for Health Information Technology (ONC) (collectively, ASTP/ONC) in response to the [Request for Information](#) (RFI) on regulatory reform on Artificial Intelligence (AI). Our members are eager to provide input to help inform your effort to identify existing Federal statutes, regulations, agency rules, guidance, forms, and administrative processes that unnecessarily hinder the development, deployment, and adoption of AI technologies, particularly in healthcare, within the United States. We strongly urge the creation of a risk-based federal policy that preempts conflicting state laws and streamlines regulations, ensuring a consistent national approach that reduces barriers and supports innovation in AI.

HLC is an association of CEOs and C-suite executives from all sectors of healthcare working to shape the future of the U.S. healthcare system. HLC is the exclusive forum for the nation's healthcare industry leaders to lead on major, sector-wide issues, generate innovative solutions to unleash private sector ingenuity, and advocate for policies to improve our nation's healthcare delivery system. Members of HLC – hospitals, academic health centers, health plans, pharmaceutical companies, medical device manufacturers, laboratories, biotech firms, health product distributors/ wholesalers, post-acute care providers, homecare providers, group purchasing organizations, and information technology companies – advocate for measures to increase the quality and efficiency of healthcare through a patient-centered approach.

The Confidentiality Coalition is a diverse alliance dedicated to balancing the protection of confidential health information with the need for efficient, interoperable systems that enhance healthcare delivery. The Coalition's aim is to safeguard the privacy of healthcare consumers while improving the essential flow of information necessary to deliver high-quality, timely, and effective care and facilitating the development of innovative medical interventions.

Overview: AI and Healthcare

For decades, AI has been a valuable tool in healthcare, recognizing patterns that help guide clinical research, care delivery, medical supply chains and more. As more sophisticated machine-learning algorithms and generative AI have emerged, human elements of healthcare can be augmented even further to serve patients and communities. AI-enabled capabilities have the potential to sharpen the accuracy of diagnoses, improve clinical decision-making, detect public health threats, and enable earlier disease intervention, with applications ranging from administrative tasks to operational efficiencies and predictive analytics for disease outbreaks. Despite these promising advancements,

significant barriers continue to impede widespread adoption and integration of AI across the healthcare ecosystem. Overcoming these barriers is crucial to unlocking AI's full potential to enhance patient care, streamline operations, and improve public health outcomes.

Barriers and Burdens to Remove

HLC and the Confidentiality Coalition strongly support the removal of unnecessarily burdensome requirements that restrict the development and deployment of AI; we share your goal of enabling the growth of private-sector AI innovation and advancements to improve health. The emerging patchwork of regulations at the federal and state level is, perhaps, the most significant threat. Private sector companies currently navigate complex and potentially conflicting regulatory and legal frameworks pertaining to data science use in healthcare and beyond. Additionally, there is uncertainty regarding best practices and the risk of a state-by-state patchwork emerging without federal guidance. The myriad requirements create confusion and inconsistencies that stifle innovation, increase compliance burdens, and hinder the seamless adoption of evidence-based AI-driven advancements – underscoring the essential need for a clear, cohesive federal framework to provide regulatory certainty.

To harness the full potential of AI, the U.S. healthcare system—public and private sectors, and regulators—must act together to enable progress. A unified federal approach to regulation is needed to harmonize expectations, reduce ambiguity, and give organizations the predictability to adopt AI responsibly.

Developed by cross-sectoral AI subject matter experts representing nearly 30 companies across the healthcare ecosystem, this report outlines barriers to uniform AI adoption in the U.S. and delivers targeted policy recommendations to address them. It urgently calls for a clear, forward-looking national standard that defines AI and establishes what responsible governance must entail.

The Report (detailed in Addendum 3) identifies three key barriers to adoption and nine policy recommendations:

BARRIER: Governance and Regulatory Complexity

Inconsistent and intricate regulations impede AI integration at scale.

Policy Recommendations:

- **Centralize Legislation:** Establish a comprehensive federal policy baseline, ensuring risk based oversight and human-in-the-loop, to harmonize and preempt conflicting state laws.
- **Modernize Regulations:** Update key regulatory frameworks such as the Health Insurance Portability and Accountability Act (HIPAA) to reflect today's digital reality.
- **Clarify Accountability:** Define accountability for AI outputs across all stakeholders.
- **Ensure Liability Protection:** Create clear legal frameworks and safe harbor provisions for responsible use of AI for patient care.



BARRIER: Data and Infrastructure

Issues surrounding data access and privacy dissuade widespread adoption.

Policy Recommendations:

- **Fortify Data Readiness, Accessibility, and Sharing:** Strengthen national standards for data quality, security, interoperability, and AI-ready data access.
- **Establish Transparency Standards:** Set federal standards for clear disclosure of AI capabilities, limitations, and ongoing monitoring.
- **Mitigate Bias:** Create federal standards to reduce bias in healthcare AI through representative data, ongoing output monitoring, and mandatory corrective action.
- **Standardize Terminology:** Develop universal terminology for what constitutes “AI use” that bridges technical and healthcare-specific contexts across research, clinical, policy, and operational domains.

BARRIER: Capabilities and End User Trust

Skill gaps and trust among users inhibit effective AI implementation.

Policy Recommendation:

- **Enhance Workforce Training and Education:** Embed AI literacy and ethics into K–12 education and upskill healthcare professionals through AI-focused education and training.

The report also outlines guiding principles and the potential benefits of responsible AI regulation.

By addressing the key challenges identified and acting on the report’s proposed recommendations, federal policymakers can catalyze the healthcare industry to unlock AI’s full potential. Removing barriers to AI adoption can improve care quality, strengthen the healthcare workforce, lower costs for American taxpayers, reduce fraud, waste, and abuse, minimize errors, and enhance patient centered care across the healthcare ecosystem.

This report aims to serve as a roadmap to inspire the necessary public-private collaboration needed to realize AI’s healthcare potential. The private sector continues to stress that it is unable to achieve the full benefits of AI without public sector collaboration. From the nearly 30 voices from across the care continuum synthesized into the report’s findings, policymakers and other public sector AI leaders can glean actionable insights to build upon in future partnerships to responsibly and safely unleash AI.

Establishing a comprehensive federal policy that preempts, deters, and possibly places a temporary moratorium on the adoption of conflicting state laws is likely the single most effective way to address barriers to AI adoption in healthcare. A comprehensive federal policy would ensure a consistent, nationwide approach and deter the widespread adoption of a state-by-state approach that hinders innovation and is resulting in an uneven set of standards being adopted across states. At the same time, a federal framework should be flexible enough to adapt to rapid AI innovation and secure enough to reduce the unnecessarily complex aspect of a fragmented approach.

Principles To Guide Further Action

To inform your effort to develop appropriate policy frameworks to enable innovation while safeguarding the public interest, the Healthcare Leadership Council and the Confidentiality Coalition



share our principles for the responsible development and use of AI in healthcare as you consider future action, in Addendum 1.

These principles support the intent to advance AI developments in support of better healthcare. Specifically, they emphasize the need for cooperation between regulators and stakeholders to balance innovation with individual rights and patient safety. Our principles call for a federal standard that preempts state regulation, is risk-based, harmonizes privacy standards for all information used, and promotes appropriate transparency. Risk assessments should be flexible, tailored to specific use cases, and align with established frameworks such as the National Institute of Standards and Technology's (NIST) AI Risk Management Framework.

The benefits of unleashing AI can be realized by allowing innovation to enhance the roles of scientists, clinicians and caregivers; assessing applications from a risk-based approach; developing a federal uniform standard; and adhering to privacy and security protections. Further, steps to harmonize requirements across business sectors and improve agencies' collaboration, along with increasing accountability, transparency, clarity, and comprehensiveness are imperative. We urge you to consider existing regulations that apply to AI enabled solutions and to leverage critical collaborative opportunities with the private sector to enable innovation and promote benefits for developers, deployers, and ultimately patients.

Instructive Use Cases Demonstrate the Potential

As AI continues to revolutionize healthcare, our member organizations are driving innovation. A collection of sample use cases detailed in Addendum 2 provides instructive examples of private-sector AI advancements in healthcare that:

- Facilitate the access and distribution of resources,
- Improve diagnostic capabilities and sharpen prognoses,
- Enhance clinical decision-making,
- Refine research and spur therapeutic development, and
- Bolster staff efficiencies and efficacies.

These AI tools illustrate both the breadth and opportunity of private sector ingenuity in leveraging AI.

Conclusion

HLC and the Confidentiality Coalition appreciate the opportunity to respond to the RFI on regulatory reform and AI. We stand ready to partner in the effort to remove barriers to the most effective uses and widescale adoption of AI in the united goal of improving outcomes for patients. Thank you for considering our comments on this RFI.

We welcome the opportunity for continued collaboration and stand ready to leverage the expertise of our private sector members to strengthen and develop effective, forward-looking solutions. If you have any questions, please do not hesitate to contact me at ckeane@hlc.org or (202) 449-3441.

Sincerely,

Clara Keane
Senior Director of Policy



Principles for the Responsible Development and Use of Artificial Intelligence in Healthcare

Introduction

Artificial intelligence (AI) is rapidly evolving, and improving healthcare delivery, augmenting research and development across drugs and medical devices, aiding diagnoses, assisting intricate procedures, streamlining administrative tasks, and identifying individuals who could benefit from additional health services. AI's full potential in healthcare can only be achieved with collaboration between private and public sector stakeholders to foster continued innovation. Our members have collaboratively developed the principles below to guide a policy framework governing the use of AI in the healthcare field.

Principles

Benefits and Innovation. The use of AI tools in healthcare lowers costs, reduces human error, increases access to care, augments existing clinical practices, alleviates clerical and administrative burden, and improves health outcomes. For example, AI enables review and translation of mammograms thirty times faster than human review with 99% accuracy and reduces the need for unnecessary biopsies.¹ The benefits of existing tools represent only the first step in realizing AI's potential to revolutionize healthcare and patient outcomes. It is therefore essential that regulators, stakeholders, and AI experts work together to ensure any regulation or policy takes a risk-based, patient centered approach that supports and nurtures developing AI technology innovations, while recognizing industry-specific differences and needs.

Risk-based Approach. Regulatory agencies and organizations using AI should adopt a risk-based oversight approach to AI applications. Risk assessments should be tailored to use cases and should be entity-dependent assuming there is a baseline, agreed-upon floor to align with consensus-based risk management frameworks such as the AI Risk Management Framework (AI RMF) developed by the National Institute of Standards and Technology (NIST). Oversight processes adopted by organizations using AI should routinely identify potential risks, possible mitigation strategies, and recommended uses for the tool(s). Regulators should avoid imposing duplicative compliance requirements, or those that conflict with existing requirements, that create an unworkable operating environment and consider organizations that follow a framework such as the NIST AI RMF to be adopting best practices. It is important to permit flexibility in assessing and allaying risk to enable companies to tailor risk mitigation to the unique considerations of each AI use case - as opposed to prescriptive requirements to guide AI oversight.

Federal Standards. Any regulatory framework(s) for AI applications should be developed and applied at the federal level, while reducing regulatory compliance costs and creating a safer environment for long-term investment. A single national standard that preempts state laws in this area will avoid conflicting, patchwork requirements and facilitate compliance without unduly restricting innovation.

Privacy and Security. Personal information used in AI should be subject to robust privacy and security protections at the federal level. This involves adhering to existing health data privacy and security protections in the Health Insurance Portability and Accountability Act of 1996 (HIPAA) and applying equivalent protections for non-HIPAA health data, by establishing baseline privacy protections for the processing of non-HIPAA personal information that often comprise the datasets used to train, validate, and test AI models. Privacy considerations should be integrated into AI tools



from the start to reflect the importance of data minimization and use limitations. Individuals should have the right to be informed about the collection and use of their personal information, and the right to access, correct and, if feasible, delete their personal information.

Congress should establish a single national standard for the use of personal information not already subject to HIPAA that encompasses use in AI applications by entities also not regulated by HIPAA. Security safeguards, which may be based on guidelines such as those provided in the NIST Cybersecurity Framework and RMF, should protect against data breaches, data poisoning, exfiltration of models or training data and other threats that could expose the data used or alter the use, behavior, or performance of an AI application.

Harmonization. Federal agencies such as the Food and Drug Administration (FDA), the U.S. Equal Employment Opportunity Commission (EEOC), and the Federal Trade Commission (FTC) should collaborate to align the federal government's approach to the regulation of AI. This will allow organizations subject to the authority of different federal agencies to harmonize their AI implementation approach, to avoid confusion, support innovation, and achieve greater compliance.

While agencies may approach this technology from varying regulatory angles, whether in safety, privacy, consumer protection or otherwise, they should take a patient-centered approach and reach sufficient alignment so that compliance with one framework will not result in violation of, or inability to comply with, another. Failure to coordinate regulatory frameworks will not only create interpretation and compliance burdens but will slow AI development and deployment, and stifle innovation by creating a regulatory patchwork incompatible with the way health is delivered.

Any additional requirements on HIPAA-covered entities must balance the need to protect patient privacy while not adding unnecessary burden to the regulated entities. Alignment enables consistency to ensure requirements are complimentary. Additionally, conflicts between federal agencies' regulation of AI will hamper U.S efforts to lead globally in the regulation of AI. Other countries want to adopt a framework for the regulation of AI that harmonizes across business sectors and regulatory areas, rather than having to deal with discordant or conflicting requirements.

Accountability. Health organizations that use AI should determine and establish a risk-based structure of accountability that extends across its partnerships to ensure that their AI use cases are deployed in a responsible, fair, and consistent manner. This includes developing, implementing, and documenting principles, policies, procedures, and an internal collaborative governance structure and controls to oversee the development and use of AI applications. These controls should include quality control parameters for the data used as well as criteria against which the performance of the AI applications is monitored, evaluated, and re-evaluated, as needed, at regular intervals throughout the lifecycle to ensure that AI can be iteratively enhanced and adapt within the risk framework. Accountability should extend to the highest levels of management and should include key elements such as risk assessment, training, monitoring, and internal sanctions.

Transparency. Transparency is essential to build trust in AI technology. While organizations should protect their intellectual property or proprietary technology, they should also disclose when they are using AI tools in high-risk applications rather than instances where it is used for administrative or clerical purposes. Organizations should not be required to reveal the inner workings of their AI systems to the public or regulatory agencies, nor is there any benefit in doing so. The detailed disclosure of either data inputs or algorithmic processes would not be meaningful to patients, providers, or payers, would force AI developers to disclose their intellectual property or proprietary



technology, could create AI vulnerability risks, and may limit innovators' willingness to work with the already highly regulated healthcare industry on meaningful AI applications.

Explainability. Developers of AI applications for use in healthcare must be able to explain to users how a decision is made by a high-risk AI application in a way that is sufficiently understandable to those users. Users should be able to gauge the context in which an algorithm operates and understand the implications of the outcomes. Users should in turn be able to explain the role of algorithms to individuals affected by AI-assisted decisions. Explanations should be meaningful and useful, tailored to the audience, and calibrated to the level of risk.

Addressing Adverse Bias and Discrimination. There are instances where AI applications have been found to be biased, as the underlying, especially historical, data sets may lack representative or accurate data. Organizations should take comprehensive steps to identify and mitigate potential sources of harmful bias across the lifecycle of their model development, and where reasonable and appropriate for specific models, align with industry-developed standards. Ongoing inclusion of sensitive personal data or data of vulnerable groups in AI training data is critical where such data is relevant for the intended use and/or population can help alleviate and remedy biases, so all patients may benefit from the advances from AI in healthcare.



Sample Use Cases of Innovative AI Advancements in Healthcare

Access & Distribution		
Company	Link	Description
Cardinal Health	Inside Cardinal's new AI-Driven Warehouse & Fulfillment Center	Cardinal is combining two facilities into one single distribution and fulfillment center, wherein robotics and technological systems will coordinate processes such as inbound/outbound and automatic storage.
Johnson & Johnson	Six ways Johnson & Johnson is using AI to help advance healthcare	J&J is using AI models to accurately predict supply and demand while identifying and responding to various risks that can impact J&J's ability to manufacture and deliver products to patients. These algorithms analyze historical data on patient demographics, demand fluctuations and supplier performance, along with real-time events that may disrupt the supply chain, to ensure customer needs are met with reliable supply.
Mount Sinai	Assessing calibration and bias of a deployed machine learning malnutrition prediction model within a large healthcare system	Mount Sinai studied its MUST-plus to evaluate current calibration and bias at the time and gave itself recommendations to improve thereafter.

Improved Diagnoses/Prognoses		
Company	Link	Description
Epic	Exploring Generative AI Adoption	Outpatient note summarization helps clinicians quickly review recent changes for a patient prior to an outpatient visit. UMCG found the AI summaries to be comparable to physician-created summaries in terms of completeness, correctness, and trustworthiness, but it took the physicians seven minutes to write them on average while it took AI just over 15 seconds.



Johnson & Johnson	<u>Six ways Johnson & Johnson is using AI to help advance healthcare</u>	Driving greater precision in measuring disease severity and treatment response: J&J is applying AI/ML to data sets – from digital images to videos to sound recordings – to identify novel biomarkers and clinical endpoints that can help us develop and deliver the right medicines for the right patients at the right time – across oncology, immunology and neuroscience.
Mayo Clinic	<u>Tomorrow's Cure: AI Powered Stethoscopes Improve Access to Heart Care</u>	The AI Powered Stethoscope captures electrocardiogram data and heart sounds to lower the diagnostic gap in detecting peripartum cardiomyopathy and provide a safe noninvasive screening test.
	<u>Mayo Clinic accelerates personalized medicine through foundation models with Microsoft Research and Cerebras Systems</u>	In a multi-partnership, foundation models (pre-trained AI models amendable to a wide array of tasks) are being used for a genomic-purpose, which is to personalize care such that an AI model can know things to specifically look individual-to-individual based on risk.
Mount Sinai	<u>AI-enhanced algorithm improves diagnosis of REM sleep behavior disorder</u>	Sleep studies have been previously limited by the necessity of research-grade 3D cameras, but an automated machine learning method can analyze video recordings with just a 2D camera.
	<u>Artificial Intelligence May Expedite Prognoses in Cancer Care</u>	AI can deduce information by image alone, helping to profile cancer in ways that highlight high-risk features and probability of metastasis.

Enhancing Clinical Decision-Making and Efficiency		
Company	Link	Description
	<u>Amazon and Memorial Sloan Kettering Cancer Center leap toward AI-driven drug discovery</u>	A partnership sees Memorial Sloan running large language models through Amazon Bedrock and SageMaker to amplify data quality, which will allow for better monitoring of cancer status in patients as well as shrink the time required for the early stages of drug development.



Amazon	<u>AI 'frees up time for patient care,' says Amazon Pharmacy's Vin Gupta</u>	Amazon Pharmacy's goal is to eliminate menial work for doctors and clinicians such that they can focus more fully on patient interaction and counseling.
Epic	<u>Mayo AI Message Responses</u>	In Basket Art (short for Augmented Response Technology) automatically drafts text for a potential response to the patient's medical advice request MyChart message based on the text of the message and information from the patient's record. Initial pilots at Mayo Clinic showed that In Basket Art saves nurses around 30 seconds per message and drafts more empathetic responses.
	<u>Ambient Listening Helps Primary Care Doctors Finish Notes Faster</u>	Ambient listening technology documents notes and queues up orders to improve the overall patient-physician experience. Third-party ambient listening vendors integrate with Epic so that doctors can spend less time on documentation, have fewer feelings of burnout, and refocus on the vital human component of patient-centered care.

Enhancing Clinical Decision-Making and Efficiency (continued)		
Johnson & Johnson	<u>Six ways Johnson & Johnson is using AI to help advance healthcare</u>	Harnessing AI for electro-anatomical mapping of the heart: During catheter ablation procedures for atrial fibrillation, it is critical for electrophysiologists to “see” inside the heart. J&J's CARTO™ 3 System – the leading 3D heart mapping system features AI to reconstruct the left atrial anatomy of the heart. By eliminating the need for manual contouring, the system represents a new approach to anatomy creation, improving the efficiency of the procedure workflow.
Mount Sinai	<u>AI Can Reliably Predict Efficacy of Immune Checkpoint Inhibitor [(ICI)] Therapy</u>	Machine learning model SCORPIO “remains a highly accessible model for predicting ICI efficacy and can aid clinical decision-making when use alongside other assessments.”
Mount Sinai	<u>Mount Sinai awarded the Hearst Health Prize for an AI-driven nutrition application</u>	AI-driven nutrition application NutriScan AI more quickly detect signs of malnutrition in hospitalized patients and informs treatment for them.
	<u>Implementing a Machine Learning (ML) Screening Tool for Malnutrition: Insights</u>	ML tool Malnutrition Universal Screening Tool (MUST), when properly implemented, can serve as a key example of effective



	from Qualitative Research Applicable to Other ML-Based Clinical Decision Support Systems	ML-based clinical decision support systems (CDSS), highlighting a switch from older CDSS to the most modern ML approach.
Oracle	Oracle Health Clinical AI Agent (CAA)	Oracle Health's AI-powered voice recognition technology records key elements of the physician-patient encounter to interpret the information, accurately inputs a draft note into the Oracle Health EHR, and enables the physician to quickly review and approve the clinical documentation produced.
Senior Helpers	Richland business leans on AI, tech to keep seniors in their own homes Tri-Cities Area Journal of Business	flexHome utilizes smart sensors to build pattern recognition on a senior patient/client's activity such that AI will alert family or the needed authorities if a deviation indicates a senior is having a medical emergency like a fall.
Vizient	Augmedix Awarded Vizient Contract for Ambient AI Documentation Solutions	Clients of Vizient will soon be able to access Augmedix services for drafting medical notes, meant to alleviate menial tasks and emphasize patient interaction and counseling.

Research & Development		
Company	Link	Description
Merck	AI optimizes the quest for small molecules - Merck.com	In the lead optimization phase of drug discovery, the use of machine learning accelerates the data consensus on things like all possible combinations of molecules such that new possible drugs are found sooner and pushed forward more quickly.

Staff Efficiency/Efficacy		
Company	Link	Description
	Revolutionizing Real-World Evidence: How	Generative AI allows medical evidence to be sourced outside of just clinical trials to provide timely, relevant, and necessary data to clinicians and doctors.



Amazon	Generative AI Can Simplify Data Exploration	
	Amazon Pharmacy Improved Prescription Processing Time by 90% with Gen AI	A HIPAA-compliant digital platform uses generative AI for data entry and validation which speeds the process of approving and assigning doctor-approved prescriptions to the pharmacy counter.
	Five ways Amazon Pharmacy uses AI to make getting medications more convenient for customers	Alongside improving prescription processing time, AI can forecast demand, provide real-time insurance estimates, answer questions ahead of an agent and conduct batching decisions.
Envision Healthcare	How Envision Healthcare Is Incorporating AI into Radiology Workflow Healthcare Innovation	By partnering with Aidoc, Envision Healthcare models the radiology workflow after the Amazon service model where ED physicians may order a study through on-site systems that will more promptly monitor a patient's case.

Staff Efficiency/Efficacy (continued)		
Company	Link	Description
Johnson & Johnson	Six ways Johnson & Johnson is using AI to help advance healthcare	Analyzing the operating room for efficiency and physician learning: AI automates snippets of surgical video to speed up the process of surgeons reviewing their work and adjusting accordingly.
Mayo Clinic	Mayo Clinic launches Mayo Clinic Digital Pathology to modernize pathology, speed medical breakthroughs	With the help of AI-gnostics, Mayo Clinic data and expertise is being combined with machine learning to shift pathology away from analogy processes to digital ones.



Surescripts	<u>Touchless Prior Authorization Remove manual work & reach determinations faster</u>	Clinical determinations regarding prior authorization of medications can be automated when AI is fed an electronic medical record within the purviews of the pharmacy benefit manager.
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Addendum #3:

Unleashing AI's Potential for Patients: A Cross-Sectoral Roadmap for Healthcare



January 2026

Unleashing AI's Potential for Patients:

A Cross-Sectoral Roadmap for Healthcare

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Unleashing AI's Potential for Patients: A Cross-Sectoral Roadmap for Healthcare

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Foreword



Bill Coyle

*Chairperson,
ZS*



Maria Ghazal

*President and Chief Executive Officer,
Healthcare Leadership Council*

About ZS:

ZS is a management consulting and technology firm that partners with companies to improve life and how we live it. We transform ideas into impact by bringing together data, science, technology and human ingenuity to deliver better outcomes for all. Founded in 1983, ZS has more than 13,000 employees in over 35 offices worldwide. To learn more, visit www.zs.com or follow us on [LinkedIn](#).

About HLC:

The Healthcare Leadership Council is the exclusive forum for the nation's healthcare industry leaders to discuss major, sector-wide issues, generate innovative solutions to unleash private sector ingenuity, and advocate for policies to improve our nation's healthcare delivery system. Learn more about HLC's policy priorities on our website at www.hlc.org and follow us on [LinkedIn](#) and [X](#).



Executive Summary

Artificial intelligence (AI) has been revolutionizing the U.S. healthcare delivery system for decades. As the technology has evolved and become more attainable, interest in its use to advance patient care and relieve provider burdens has only increased. AI is already enhancing diagnostics and clinical decision-making, streamlining operations, expanding access to care, and accelerating drug discovery - and innovation continues at a rapid pace.

However, despite growing momentum and interest in its potential, adoption of AI across the healthcare ecosystem is inconsistent and inhibited. The current landscape is marked by evolving, disparate federal and state policy frameworks, regulatory uncertainty, infrastructure development needs, challenges in data interoperability, and growing concerns around fairness and trust. These barriers continue to stunt progress and impede the sector from achieving the full benefits of further implementation.

To harness the full potential of AI, the U.S. healthcare system—public and private sectors, and regulators—must act together to enable progress. A unified federal approach to regulation is needed to harmonize expectations, reduce ambiguity, and give organizations the predictability to adopt AI responsibly.

Developed by cross-sectoral AI subject matter experts representing nearly 30 companies across the healthcare ecosystem, this report outlines barriers to uniform AI adoption in the U.S. and delivers targeted policy recommendations to address them. It urgently calls for a clear, forward-looking national standard that defines AI and establishes what responsible governance must entail.

The report identifies three key barriers to adoption and nine policy recommendations:

BARRIER:

Governance and Regulatory Complexity

Inconsistent and intricate regulations impede AI integration at scale.

Policy Recommendations:

1. **Centralize Legislation:** Establish a comprehensive federal policy baseline, ensuring risk-based oversight and human-in-the-loop, to harmonize and preempt conflicting state laws.
2. **Modernize Regulations:** Update key regulatory frameworks such as the Health Insurance Portability and Accountability Act (HIPAA) to reflect today's digital reality.
3. **Clarify Accountability:** Define accountability for AI outputs across all stakeholders.
4. **Ensure Liability Protection:** Create clear legal frameworks and safe harbor provisions for responsible use of AI for patient care.



BARRIER:

Data and Infrastructure

Issues surrounding data access and privacy dissuade widespread adoption.

Policy Recommendations:

1. **Fortify Data Readiness, Accessibility, and Sharing:** Strengthen national standards for data quality, security, interoperability, and AI-ready data access.
2. **Establish Transparency Standards:** Set federal standards for clear disclosure of AI capabilities, limitations, and ongoing monitoring.
3. **Mitigate Bias:** Create federal standards to reduce bias in healthcare AI through representative data, ongoing output monitoring, and mandatory corrective action.
4. **Standardize Terminology:** Develop universal terminology for what constitutes “AI use” that bridges technical and healthcare-specific contexts across research, clinical, policy, and operational domains.

BARRIER:

Capabilities and End User Trust

Skill gaps and trust among users inhibit effective AI implementation.

Policy Recommendation:

- **Enhance Workforce Training and Education:** Embed AI literacy and ethics into K–12 education and upskill healthcare professionals through AI-focused education and training.

To support execution, the report offers 25 actionable tactics, organized by level of impact and effort required, offering policymakers a clear roadmap from easier to more challenging steps.

As displayed in the report’s prioritization framework, the three most high-impact, low-effort tactics are:

1. Evaluate Existing Federal Regulations
2. Define Healthcare-Specific AI Terminology
3. Establish Guidelines for Data Cleansing & Standardize Intake

The report also outlines guiding principles and the potential benefits of responsible AI regulation.

By addressing the key challenges identified and acting on the report’s proposed recommendations, federal policymakers can catalyze the healthcare industry to unlock AI’s full potential. Removing barriers to AI adoption can improve care quality, strengthen the healthcare workforce, lower costs for American taxpayers, reduce fraud, waste, and abuse, minimize errors, and enhance patient-centered care across the healthcare ecosystem.

This report aims to serve as a roadmap to inspire the necessary public-private collaboration needed to realize AI’s healthcare potential. The private sector continues to stress that it is unable to achieve



the full benefits of AI without public sector collaboration. From the nearly 30 voices from across the care continuum synthesized into the report's findings, policymakers and other public sector AI leaders can glean actionable insights to build upon in future partnerships to responsibly and safely unleash AI.

The nine policy recommendations outlined in this report represent Healthcare Leadership Council's core proposals, while the 25 accompanying actionable tactics are illustrative, non-prescriptive pathways intended to demonstrate how the recommendations could be advanced. Examples are included to illustrate approaches and initiatives currently observed across the healthcare landscape. Their inclusion does not imply endorsement, opposition, or a unified position by HLC, its members, or the authors of this report.

Purpose, Call to Action, and Guiding Principles

This report offers recommendations to federal legislative and regulatory efforts focused on fostering responsible, unbiased, and scalable AI adoption in healthcare. Drawing from nearly 30 members of HLC, existing standards, and past precedence, our goal is to equip lawmakers and agency leaders with actionable strategies to reduce barriers and accelerate AI integration.

By embracing AI technologies and unlocking the full potential of available data, the U.S. has the opportunity to:

- **Improve patient experiences and expand access** through faster, scaled, more accurate, and personalized care that improves health outcomes, well-being, and access to care.
- **Reduce the cost burden on U.S. taxpayers** through greater efficiency and smarter resource allocation.
- **Alleviate healthcare workforce shortages and burnout** by augmenting—not replacing—human expertise.
- **Decrease healthcare-related fraud, waste, and abuse** with advanced detection and prevention tools.
- **Reduce the frequency of mistakes and errors** to enhance safety and quality of care.

The sector is already proving the power of this technology, as evidenced in many use cases, such as:

- In 2025, a multicenter, double-blind, randomized, placebo-controlled Phase 2a trial of a first-in-class, [generative AI-discovered TNIK inhibitor](#) for idiopathic pulmonary fibrosis was completed. Beyond meeting its primary safety endpoint, the study reported improvement in lung function at the highest dose, highlighting that an AI-generated compound can produce patient-relevant signals in humans and merits larger trials.
- A [prospective, multi-site study](#) (over 590k encounters across five hospitals) of Johns Hopkins' TREWS AI-enabled early-warning system for sepsis found that when clinicians confirmed an alert within 3 hours, patients had an 18.7% relative reduction in in-hospital mortality, along with less organ failure and shorter stays.



- Ambient tools directly address one of clinicians' biggest barriers to AI adoption, administrative overload, and free up time for patient care while maintaining clinician oversight of notes. A [2025 JAMA Network Open pre-post study](#) across 17 specialties using Nuance DAX (ambient clinical intelligence) reported 20.4% less time in notes per appointment, 30% less after-hours work per day, and higher same-day note closure.

The following five guiding principles are essential to address adoption barriers and inform key policy recommendations:

Leverage public-private partnerships.	Prioritize collaboration between government and industry to establish clear regulatory guardrails and ensure consistent enforcement of federal AI policies in healthcare. Government sets clear guardrails for safety and effectiveness; private sector partners provide practical, real-world insights for implementation, ensuring regulations are actionable and aligned with healthcare realities.
Establish balanced regulations.	Promote AI innovation while ensuring robust protections are in place to mitigate risks and prevent harm to patients. Provide privacy guardrails and regulatory clarity without stifling innovation, adding unnecessary administrative burden, or delaying improvements in care.
Enable agility for AI evolution.	Ensure policies and national standards are flexible enough to adapt to rapid AI advances while maintaining predictability for developers, healthcare providers, and patients.
Ensure human oversight and accountability.	Require that AI complements rather than replaces human judgment by integrating human oversight throughout the AI lifecycle. Foster a culture of co-intelligence that ensures humans are in the loop.
Align to a risk-based approach.	A risk-based framework must classify AI solutions in healthcare based on their potential impact and tailor oversight accordingly. Not all AI is the same and should therefore be evaluated based on the intended outcome or purpose of the AI.



Barriers to AI Adoption at Scale

While AI holds immense promise to transform healthcare, real-world adoption faces significant, industry-specific hurdles. In-depth interviews with experts across every sector of the healthcare ecosystem point to three interdependent areas where progress can benefit from further clarification and/or support:



Governance and Regulatory Complexity



Data and Infrastructure



Capabilities and End User Trust



Governance and Regulation Complexity

Inconsistent and intricate regulations impede AI integration at scale.

- **Regulatory Fragmentation:**
Healthcare organizations must navigate inconsistent, and often conflicting, state, federal, and international AI regulations. Without a unified federal framework, compliance becomes costly, burdensome, and uncertain—especially for entities operating across multiple jurisdictions.
- **Pace of Innovation vs. Regulatory Lag:**
AI advances faster than regulatory policy can adapt. Slow and complex federal policymaking leaves innovators unsure whether emerging technologies and techniques will comply with outdated or unclear rules, discouraging investment.
- **Bias and Ethical Risks:**
AI models trained on incomplete or non-representative datasets risk perpetuating bias and discrimination. Standardized federal requirements for demonstrating fairness, representativeness, and accuracy are needed to maintain patient wellbeing.
- **Accountability and Legal Ambiguity:**
Uncertainty regarding liability and legal precedent complicates the assignment of fault when AI systems fail or cause harm. The lack of clarity—across developers, data providers, clinicians, manufacturers, and users—complicates business agreements, malpractice coverage, and legal risk management.





Data and Infrastructure

Issues surrounding data access and privacy dissuade widespread adoption.

- **Data Privacy and Security:**

The integration of AI in healthcare, particularly concerning patient outcomes, faces significant challenges related to data privacy, patient consent, and the complex global regulatory landscape. While HIPAA provides essential protections for certain information when used by covered entities, the inconsistencies in state laws, the varied nature and volume of data collected, and sometimes, international regulations like the [EU's General Data Protection Regulation](#) create hurdles for data access and usability. Additionally, threats to data security, particularly in today's digital landscape, heighten risks as data protected within healthcare institutions can be compromised when accessed through less secure third-party applications. Without national standards for consent and privacy, innovation in healthcare may stagnate.
- **Data Access:**

There are many factors that contribute data extraction for AI applications including but not limited to: differences in technological maturity, inconsistent data codification, and purpose-based standards. Much of this data remains siloed, locked behind paywalls, or technically incompatible, often increasing the cost of extracting and cleaning data. For AI to be developed for purposes such as diagnosing, surfacing incidental radiology findings, or recommending treatment protocols, training sets must be diverse, representative, and comprehensive. Enforceable data standards and accessible infrastructure are essential to realizing AI's full potential.
- **Interoperability and Integration:**

Despite progress, current efforts to support AI tools within healthcare continue to face challenges with interoperability. The incomplete adoption of consistent standards for data extraction and use are limiting the ability of AI applications to meaningfully interoperate with electronic health records, imaging platforms, and other clinical systems. This results in fragmented data exchange and inconsistent performance across and within health systems.
- **Cost, Resource, and Infrastructure Limitations:**

The high costs of implementing, safeguarding, and maintaining AI solutions create barriers, especially for underserved organizations. Significant investments in technology infrastructure, data procurement, and skilled personnel are often required, hindering broader adoption. Additionally, many providers face challenges integrating AI due to outdated systems and limited resource capacity.



Capabilities and End User Trust

Skill gaps and trust among users inhibit effective AI implementation.

- **Domain Expertise Deficit:**

A critical barrier to effective AI adoption in healthcare is the lack of medical expertise among most AI developers. Without a strong understanding of clinical practice, developers are not well equipped to validate or train AI systems for real-world medical use. This gap often forces



healthcare organizations to divert clinical staff, such as nurses, from direct patient care to assist in AI training and implementation.

This not only strains resources amid nationwide workforce shortages but also increases the risk that AI tools fail to meet clinical standards. Where none already exist, targeted initiatives to embed medical expertise into AI development, coupled with structured training for implementation teams, are essential to ensure AI solutions are safe, effective, and aligned with patient care priorities.

- **Workforce Resistance and Patient Trust:**

There is some lack of trust and comfort with AI among healthcare workers and patients. Many practitioners remain skeptical due to fears of job displacement and potential legal liability in case of AI-errors. These concerns are compounded by the “black box” nature of many AI systems, including concerns around the completeness of the training data and limited visibility into how models are built and how outputs are generated.

Patients also share concerns, particularly around the accuracy of AI outputs and the protection of their personal health data but tend to be less skeptical than providers. Past data breaches, the commercialization of health apps, and uncertainty about how AI informs clinical decisions reinforce the demand for greater transparency, accountability, and oversight. An additional significant risk is the tendency for end users to over trust AI outputs, assuming they are always correct. This overreliance can lead to serious errors when decisions are made without human judgment or critical evaluation.



Policy Recommendations to Break Down Barriers

By embracing AI technologies and unlocking the full potential of available data, the U.S. can improve overall patient outcomes across communities, alleviate workforce shortages, reduce costs for taxpayers, detect and curb healthcare-related fraud, and reduce the burden of mistakes and errors.

This section presents a series of nine policy recommendations designed to address key barriers to AI adoption in healthcare. Each recommendation is grounded in the guiding principles outlined above and mapped out to one of three previously articulated categories of barriers (i.e. governance and regulatory complexity, data and infrastructure, and capabilities and end user trust).

Each recommendation includes actionable tactics (25 total) detailing practical potential next steps for addressing these opportunities. These recommendations are supported by examples that demonstrate precedent and are intended to be used as general learnings that regulators can take or adopt relevant attributes where applicable. These examples are neither endorsed nor opposed by HLC, its members, or the authors of this report.

Under each recommendation, a scale for the actionable tactics serves as a guide to portray the relative level of effort and feasibility to implement the particular recommendation. The left, white section is reflective of a tactic that is easy to implement whereas, the right, red section is reflective of a tactic that would be challenging to implement.

Note: The following recommendations are paired with the barriers they would address and are not represented in order of priority or importance.

Page 25 features a heat map that organizes each actionable tactic by its potential to drive meaningful change (impact) and the feasibility (effort) of implementing it.





BARRIER

Governance and Regulatory Complexity

Inconsistent and intricate regulations impede AI integration at scale.

Recommendation: Centralize Legislation

Establish a comprehensive federal policy that preempts conflicting state laws to ensure a consistent, nationwide approach to mitigating AI-related risks in healthcare. This framework should define currently unaddressed areas—such as permissible levels of data de-identification and required safeguards for compliance—while remaining flexible enough to adapt to rapid AI innovation. A single federal standard will reduce regulatory fragmentation and foster responsible adoption.

TACTICS

Review Current State Legislative Landscape:

Review existing state AI-related laws to identify effective practices, inconsistencies, and gaps, and assess their impact on healthcare. The U.S. has several instances that can be leveraged as a starting point, but we can also learn (wins and potential pitfalls) from international regulations that are well ahead on this path to standardization.

- U.S. Example: [California's AI Risk Management Framework](#), linked to the California Consumer Privacy Act and California Privacy Rights Act introduces transparency requirements for automated decision-making. Other states, such as [Colorado \(SB21-169\)](#), address algorithmic bias in specific sectors (e.g., insurance) but lack comprehensive healthcare AI provisions—highlighting the need for federal alignment.
- International Example: The [EU AI Act](#) uses a risk-tiered model, classifying certain health-related AI as “high-risk” with strict oversight and documentation requirements. While this framework demonstrates the commitment to protections, its complexity and potential overreach offer learnings for US policy makers to be aware of. In contrast, the [UK's MHRA \(Medicines and Healthcare products Regulatory Agency\)](#) “regulatory sandbox” allows controlled real-world testing—offering a potential model for U.S. pilot programs.

Establish a State Law Moratorium:

Rather than a sweeping ban, a federally guided, narrowly scoped pause on state laws related to use of AI in healthcare should be considered.

- Example: Although the proposed five-year moratorium on state-level AI enforcement in the [One Big Beautiful Bill Act](#) was removed in a nearly unanimous 99-1 Senate vote in early July 2025 after clearing the House, its consideration highlights bipartisan concerns about an inconsistent state patchwork as well as a broad and lengthy preemption of state law.
- Example: In December 2025, President Donald J. Trump signed an [executive order](#) to block states from implementing excessive regulations and create a patchwork of varying



"regulatory regimes." Through the development of a framework serving as a "minimally burdensome national standard," the executive order is intended to cultivate an environment for U.S. AI companies that enables freedom in innovation.

Develop a Comprehensive Federal Policy:

Collaborate with private sector entities (across healthcare and technology) to shape a comprehensive federal policy framework that incorporates all the aspects of successful AI development, deployment, adoption, and scaling. This framework should include data privacy standards, evaluation and safety standards, post-market monitoring protocols, and transparency guidelines. HLC members widely agree that a unified set of regulations would simplify compliance and ensure consistency across states.

- Example: While not enforceable regulation, nor tailored for healthcare, the National Institute of Standards and Technology's (NIST's) [AI Risk Management Framework](#) (RMF 2.0) outlines voluntary standards for AI governance, addressing key areas such as data integrity, system safety, transparency, and accountability. It exemplifies how public and private sector collaboration can produce practical, consensus-driven guidance that balances innovation with risk management.

Relative Ease of Implementing Actionable Tactics to Advance the "Centralize Legislation" Recommendation



Recommendation: Modernize Regulations

To keep pace with the rapidly evolving digital and AI landscape, federal regulations must be modernized to integrate within outdated frameworks that limit responsible scaling and widespread adoption. Updating core policies—such as HIPAA and others—will foster accountability, consistency, and agility, enabling healthcare organizations to innovate without being constrained by legacy policies and standards. Modernized regulations should provide clear, forward-looking guidance on critical data privacy and security challenges, including consent management, uniform security requirements, and protections aligned to today's digital reality.



TACTICS

Evaluate Existing Federal Regulations:

Assess current regulatory standards to identify gaps in applicability, coverage, and enforcement—particularly where definitions (e.g., protected health information, de-identification) no longer reflect today’s technology and data practices.

Revise and Modernize Frameworks:

Update existing regulations, including HIPAA, to address accountability, data governance, security and provenance, while retaining the flexibility provided by HIPAA’s permitted uses of data for treatment, payment, and health care operations. While existing ONC HTI-1 standards promote interoperability and foundational data transparency, the Health IT framework must go further to address fairness, accountability, and real-world safety for widespread adoption. This framework should clarify display requirements and coverage expectations to ensure all providers delivering health AI solutions are accounted for—not just a subset. We also support utilization of regulatory sandboxes to test approaches and adjust to enable the unified federal approach to regulation and requirements.

- Example: While [HIPAA](#) provides strong federal standards for protected health information (PHI), it has not received a major update since 2013. Its current security standards do not fully address the realities of AI-driven healthcare, leaving gaps in areas such as consent, risk classification, and accountability. Updating HIPAA to reflect evolving technologies would strengthen privacy, security, and trust in an AI-rich environment.
- Example: FHIR (Fast Healthcare Interoperability Resources) was originally designed to provide a set of rules and specifications for the secure exchange of electronic healthcare data, but initiatives like SMART (Substitutable Medical Applications, Reusable Technologies) on FHIR, AI Metadata Extensions, and Clinical Decision Support (CDS) hooks are being introduced and piloted across use cases to expand coverage to AI workflows. The pace of establishment, adoption, and enforcement of these standards will support keeping up with the rapid pace of demand, innovation, and utility of AI within the healthcare ecosystem to formalize the process.

Relative Ease of Implementing Actionable Tactics to Advance the “Modernize Regulations” Recommendation



Recommendation: Clarify Accountability

Clearly define and assign accountability for AI outputs across all parties, including solution developers, data and technology providers, and healthcare providers, to eliminate ambiguity in liability and build trust in AI-enabled care. Current healthcare regulations and standards do not extend to most healthcare value chain partners or developers, as well as healthcare service providers, leaving significant accountability gaps.

TACTICS

Documentation of Accountability:

Establish federal requirements that clearly define roles, responsibilities, and oversight for all parties involved in developing, deploying, and monitoring AI systems, ensuring patients and caregivers understand who is responsible at each stage.

- Example: The [EU AI Act](#) requires clear documentation of roles and responsibilities for “high-risk” healthcare AI, with providers accountable for design and validation and deployers responsible for data quality, oversight, and communication of system limitations. As U.S. policymakers consider similar approaches, it will be important to evaluate how the EU AI Act has impacted AI innovation in the EU, and whether this model, or an alternative one, would best align with the U.S. healthcare landscape. We note that HIPAA applies only to covered entities, leaving many AI developers, digital health platforms, and third-party aggregators outside federal privacy and security obligations despite handling sensitive health data.
- Example: In early 2025, the FDA issued a [Draft Guidance](#) on “Considerations for the Use of Artificial Intelligence to Support Regulatory Decision-Making for Drug and Biological Products.” This guidance incorporates risk-based considerations and may serve as a starting point for developing future regulatory frameworks and best practices for AI in healthcare.

Establish Standards for Non-covered Entities:

Develop comparable [HIPAA](#)-aligned standards to ensure the privacy and security of personal health information and health-related data are protected when used by non-covered entities. Today, many AI developers, digital health platforms, and third-party aggregators may be subject to different federal or state privacy and security obligations despite their significant role in the handling of sensitive health data.

Expand Malpractice Insurance Coverage:

Revise existing malpractice insurance statutes to account for AI-enabled clinical decision-making and clearly distinguish when liability lies with clinicians (e.g. negligent use), healthcare institutions (e.g. insufficient controls), or technology partners (e.g. poor modeling), to protect responsible users while reinforcing oversight and safety.



Relative Ease of Implementing Actionable Tactics to Advance the “Clarify accountability” Recommendation



Recommendation: Ensure Liability Protection

To foster responsible AI innovation, federal policy must clearly address liability in the use of AI for patient care. This includes defining permitted uses of AI in patient care, setting expectations for controls, and outlining protections for organizations that comply with established standards and best practices. Reducing liability uncertainty will encourage more developers and healthcare organizations to design, deploy, and scale AI solutions safely.

TACTICS

Establish and Incentivize Industry Certifications and Audits:

Create auditable standards, certified by recognized industry bodies to promote adoption of safe development practices.

- Example: The [HITRUST AI Assurance Program](#) offers AI Security and Risk Management Assessments to help organizations secure AI-powered applications and demonstrate trustworthiness. Establish mechanisms that promote universal adoption of this or an equivalent assessment, making healthcare organizations responsible for ongoing certification and monitoring.

Develop a Federal Legal Framework:

Provide clear guidance on interpretation, culpability, burden of liability, and permitted use across the spectrum of healthcare AI applications. Several interviewees, especially data partners, suggest that federal leadership should set national standards, convene industry partners, and address liability to enable safe and effective AI use.



Establish a Regulatory Safe Harbor Provision:

Offer targeted liability protection for organizations that meet defined guardrails, transparency, and monitoring requirements.

- Example: A modernized “[Patient Engagement and Support Safe Harbor](#)” could expand protections to additional entities and permitted support areas, enabling broader care coordination while maintaining safeguards.

Relative Ease of Implementing Actionable Tactics to Advance the “Ensure Liability Protection” Recommendation



BARRIER

Data and Infrastructure

Issues surrounding data access and privacy dissuade widespread adoption.

Recommendation:

Fortify Data Readiness, Accessibility & Sharing

Strengthen data quality, security, and interoperability standards to ensure healthcare information is secure, accessible, portable, and ready for AI integration throughout its lifecycle. This includes providing clear federal guidance on data access, sharing, usage, and security tailored to AI applications across healthcare. Mandating standardized digital information sharing for payment and operations use cases will help reduce delays, denials, and invalid claim edits that negatively impact both patients and providers.

Anchored in FHIR and the U.S. Core Data for Interoperability (USCDI) standards, the National Coordinator for Health Information Technology’s (ONC’s) Cures Act Final Rule establishes nationwide requirements for the standardized exchange and accessible sharing of electronic health record (EHR) data. These rules enable secure connections between third-party AI tools and patient information while emphasizing the need for structured, high-quality data to ensure both regulatory compliance and trust across the healthcare system.



TACTICS

Establish Guidelines for Data Cleansing and Standardize Intake:

Develop comprehensive protocols that outline the process for identifying, correcting, and removing inaccurate or irrelevant data. Also establish standards for how data should be collected, formatted, and input to ensure consistency and accuracy across all datasets to improve future use.

Promote Integration, Interoperability, and Standards-based Exchange:

Broadly incorporated by major EHR vendors and providers, ONC's FHIR and USCDI standards represent meaningful progress toward nationwide interoperability. Despite this progress, true data portability and seamless exchange remain challenging and costly, particularly for small providers and practices with limited resources. FHIR implementations vary as vendors often support multiple FHIR versions in parallel, and data beyond the standard US Core profiles can include proprietary extensions, which can lead to 'FHIR profiles' that complicate uniform integration. As the list of clinical information required by USCDI continues to expand, these certification-required capabilities should take into account what data elements are most important for AI. Beyond these technical barriers, the cost and resources (i.e. people and expertise) required for implementation and ongoing updates is high; many smaller providers and practices struggle to comply in a timely manner. As a result, while EHR developers have advanced significantly within their domain, factors beyond their direct control continue to limit widespread and uniform adoption. Federal leadership and targeted incentives will be essential to close these gaps, reduce burdens on smaller organizations, and deliver on the promise of true interoperability across the healthcare system.

Establish a Patient Data Registry:

Develop a national patient registry with a digital consent ledger, inclusive of an opt-out control, that enables patients to track where their data resides, control who can access it, and manage permissions to determine their data sharing preferences. As suggested in the interviews, by many data brokers, manufacturers, and providers, a patient ledger puts patients at the center, safeguarding privacy while enabling responsible data sharing to support innovation, including AI.

While precedents exist such as the oncology registries, the recently mandated Alzheimer's therapy registry, and the [Health Information Exchanges from the Definitive Healthcare](#), these efforts are fragmented and narrow in scope. They stop short of offering a comprehensive, patient-controlled system that spans the full healthcare ecosystem.

The July 2025 announcement from the current Administration to develop a patient centric healthcare ecosystem (in partnership with the private sector) marks progress in advancing aligned networks, securing digital identity, and consistent data exchange standards. This initiative lays the groundwork, but realizing true patient controlled centralized data repository will require significant funding, strong regulatory governance, as well as strong commitment from public and private sectors.



- Example: Some learning from the [UK's Biobank](#) (while still research focused), may offer a starting point for a centralized, consented dataset that allows broad reuse while maintaining participant control (not at a granular level) and transparency.

Relative Ease of Implementing Actionable Tactics to Advance the “Fortify Data Readiness, Accessibility & Sharing” Recommendation

Easy

Challenging



Recommendation: Establish Transparency Standards

Establish clear federal standards for transparency in the design, deployment, and monitoring of AI solutions in healthcare. These standards should define what information organizations must disclose—such as AI capabilities, limitations, training data sources, performance metrics, and error rates—how the disclosed information must be communicated to both clinicians and patients, and requirements for ongoing monitoring. AI-specific laws, such as those that relate to transparency and ethical use, should not conflict with existing privacy laws and should take them into account.

Regulatory frameworks should strike a balance—ensuring accountability and improving adoptions without stifling innovation and limiting the potential benefits of improving patient outcomes. Greater transparency will strengthen trust among providers, patients, and regulators, while promoting responsible AI use.

TACTICS

Develop a Configurable AI “Nutrition Label”:

Create a nationally recognized disclosure framework to communicate bias safeguards, privacy protections, and algorithm performance in a clear, consistent format. The label should be configurable to allow comprehensive detail for high-risk applications and lighter specificity for lower risk use cases, as well as adaptable to the specific parameters available for each AI tool.

- Example: [NIST's AI Risk Management Framework](#) (AI RMF) aims to foster the development of innovative approaches to address characteristics of trustworthiness, including valid and reliable, safe, secure and resilient, accountable and transparent, explainable and interpretable, privacy-enhanced, and fair, with harmful bias managed.

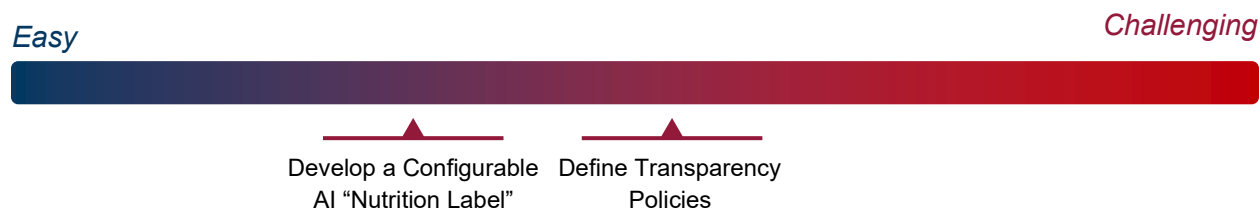


Define Transparency Policies:

Mandate disclosure of AI system performance, limitations, and risk mitigation measures in a standardized format, coupled with appropriate monitoring of outputs.

- Example: [FDA Guidance on AI-Enabled Medical Devices](#) (in draft) recommends documenting validation data, bias mitigation strategies, and limitations, and encourages the use of model cards to communicate risks and capabilities.
- Example: [ONC's HTI-1 Final Rule](#) (effective February 2024) requires algorithm transparency and risk documentation for certified AI decision-support tools, including standardized disclosure of development methods, validation results, and performance metrics.
- Example: The U.S. can learn from the [EU AI Act](#), which sets risk-tiered transparency obligations for AI applications, requiring clear instructions for use and disclosure of system capabilities, limitations, and accuracy metrics.

Relative Ease of Implementing Actionable Tactics to Advance the “Establish Transparency Standards” Recommendation



Recommendation: Mitigate Bias

Establish comprehensive federal guidelines to reduce bias in healthcare AI applications. This includes leveraging representative datasets for training, requiring appropriate monitoring of AI outputs, and mandating corrective action when biases are detected. A consistent, nationwide approach will reduce discrimination, promote fairness, and build trust in AI tools.

TACTICS

Establish Continuous Monitoring Guidelines:

Require organizations to implement periodic bias detection and transparency practices during model development and post-deployment, with mandated corrective actions to maintain fairness and accuracy.



Define AI Development Guidelines:

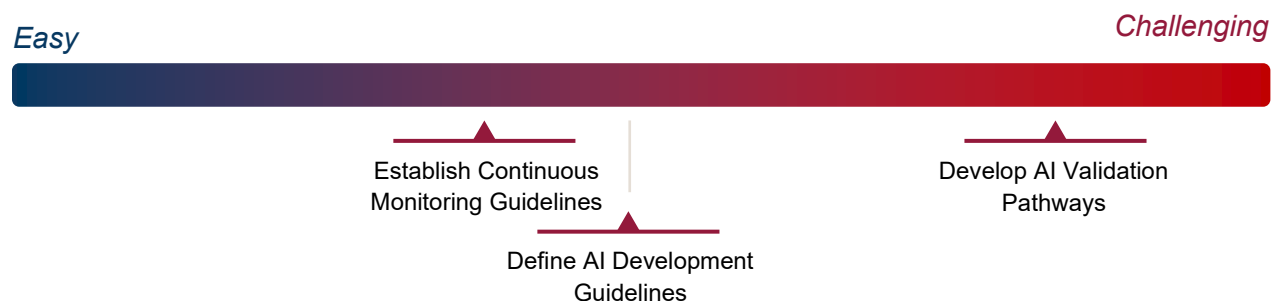
Define best practices for AI design, testing, and retraining that prioritize representative datasets and multidisciplinary development teams to more comprehensively represent multi-faceted patient populations.

Develop AI Validation Pathways:

Ensure clear validation pathways for accuracy testing, usability assessments, and simulated deployments. In line with a risk-based approach, higher risk use cases could be validated and certified by local evaluators or external experts.

- Example: A [2023 JAMA proposal](#) recommends a nationwide network of Health AI Assurance Laboratories as shared public–private resources to test and validate models against agreed best practices established by Nation Academy of Medicine.
- Example: Several health systems have developed local validation and monitoring platforms—emphasizing multidisciplinary, cross-department collaboration inside the health system—to ensure AI tools perform safely within their own clinical environments. For instance, Vanderbilt University Medical Center’s Vanderbilt [VAMOS](#) (Algorithmovigilance Monitoring and Operations System) system supports ongoing, system-wide monitoring of AI models throughout their lifecycle to promote safe and effective use. Additionally, University of California, San Francisco Health’s [IMPACC](#) (Impact Monitoring Platform for AI in Clinical Care) platform evaluates AI tools in real-world workflows and monitors performance and bias.

Relative Ease of Implementing Actionable Tactics to Advance the “Mitigate the Bias” Recommendation



Recommendation: Standardize Terminology

The definition of what constitutes ‘AI use’ in research and development remains fluid and inconsistent across the industry. As more advanced forms of AI begin to demonstrate tremendous potential for drug discovery tools, clinical decision support, care coordination, and patient engagement, the need for consistent, universal terminology becomes even more urgent. Establishing a standardized vocabulary that bridges both technical and healthcare-specific contexts will promote clarity, improve communication across stakeholders, and ensure regulations are applied consistently.

TACTICS

Define Healthcare-Specific AI Terminology:

Develop a federal glossary of AI terms tailored to the healthcare sector, like HIPAA’s definitions for health data. This glossary should include concrete and relevant examples for each of the glossary AI terms. For example, diagnostics tools in the context of medical imaging refer to algorithms that interpret medical images.

Integrate Definitions into Federal Regulatory Frameworks:

Embed the standardized glossary into federal AI regulations and related guidance to ensure uniform application across all policies and programs.

- Example: [NIST Glossary for Trustworthy AI](#) – A non-sector-specific resource for common AI and machine learning (ML) terminology.
- Example: International Organization for Standardization/International Electrotechnical Commission ([ISO/IEC 22989:2022](#)) – International consensus-based AI vocabulary widely used in governance frameworks can serve as a seed to developing a US Healthcare centric glossary.
- Example: [FDA Digital Health and AI Glossary](#) – Aligns with ISO standards and provides practical definitions for AI systems and components.

Relative Ease of Implementing Actionable Tactics to Advance the “Standardize the Terminology” Recommendation

Easy

Challenging



Define Healthcare-Specific AI Terminology

Integrate Definitions into Federal Regulatory Frameworks





BARRIER

Capabilities and End User Trust

Skill gaps and trust among users inhibit effective AI implementation.

Recommendation: Enhance Workforce Training & Education

Develop comprehensive education and training programs to upskill healthcare professionals in both foundational AI concepts and healthcare-specific applications. By embedding AI literacy into professional development, the healthcare workforce can better understand, evaluate, and responsibly adopt AI to mitigate job-loss fears among clinicians and strengthen patient trust.

TACTICS

Incentivize Employee Training:

Encourage healthcare organizations to provide workforce-wide AI training through federal financial incentives, such as tax credits or subsidies. Training should cover AI terminology, data privacy and security, transparency, bias mitigation, and healthcare use cases. This is supported by a [2022 Congressional Research Service report](#) recommendation for targeted tax credits for employer training, with an incremental model that rewards investment above a baseline and prioritizes low- and middle-wage employees to promote fair and inclusive skill-building.

- **Example:** Potentially seek to leverage the [One Big Beautiful Bill's expansion of Pell Grant eligibility](#) for short-term training programs to include AI-focused healthcare courses, enabling nurses and other mid-career providers to access affordable upskilling opportunities.

Revamp Curriculum to Drive Inherent AI-Readiness:

Adapt curricula from K–12 through postgraduate levels to include foundational AI concepts, ethical considerations, and real-world healthcare applications. AI readiness amongst the incoming workforce will ensure faster adoption and scaling across the industry.

- **Example:** The [White House Executive Order on Artificial Intelligence Education](#) published in April 2025 directs federal agencies to integrate AI literacy into K–12 curricula, train educators, and prioritize AI in discretionary grants and workforce development programs.



Establish Government Incentives:

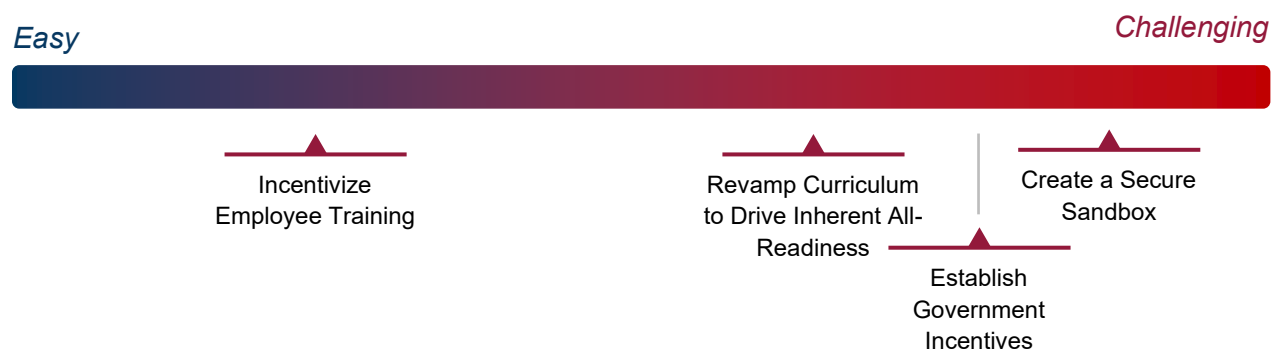
Target and direct federal incentives to under-resourced healthcare organizations and state colleges to develop and deliver AI-focused curricula, ensuring equitable access to training across geographies and roles.

Create a Secure Sandbox:

Establish controlled environments where healthcare professionals can safely test AI tools on de-identified data to further build confidence, competence, and capabilities.

- Example: The [UK's NHS "AI Airlock"](#) allows innovators to pilot AI tools in live settings under strict governance, ethical review, and patient safety oversight.

Relative Ease of Implementing Actionable Tactics to Advance the “Enhance Workforce Training and Education” Recommendation



Prioritizing Actionable Tactics to Advance Policy Recommendations

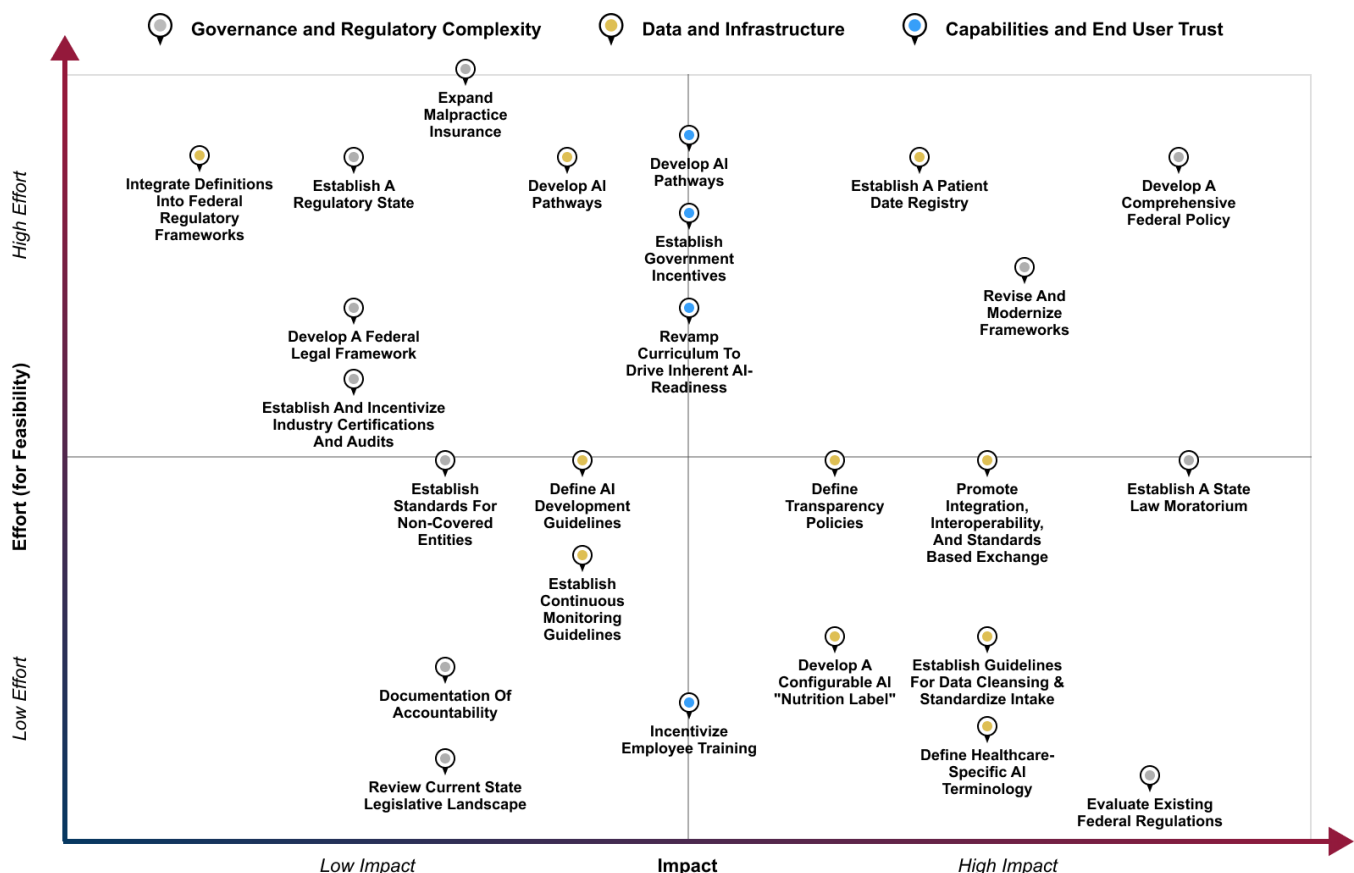
The following figure depicts a prioritization framework categorized across the three barriers:



Each actionable tactic is plotted based on its potential to drive meaningful change (i.e. impact) and the feasibility (i.e. effort) of implementing it.

This framework, informed by the cross-sector healthcare stakeholders interviewed, identifies near-term opportunities and longer-horizon investments to promote a coordinated and scalable approach to responsible AI adoption in healthcare.

Impact versus Effort for the Regulatory Actionable Tactics



Potential Benefits:

By addressing barriers, clarifying accountabilities, and investing in education and infrastructure, these actions can not only accelerate responsible AI adoption, but also ensure that the advantages of adoption are appropriately distributed across the healthcare ecosystem. A coordinated, federally guided approach can unlock the following transformative benefits across the healthcare ecosystem, with extraordinary dividends for patient care:

✓ **Improve Patient Safety and Outcomes:**

Enhance patient care and outcomes through better data quality and interoperability, ensuring AI systems are reliable and effective.

✓ **Promote AI Innovation and Adoption:**

Encourage AI innovation by providing predictability, reducing hesitation to invest, and incentivizing organizations to deploy AI with protection.

✓ **Enhance Public Trust:**

Increase public trust by promoting accountability and consistency in AI standards, while encouraging collaboration among regulators, developers, and users.

✓ **Reduce Compliance Burden and Legal Uncertainties:**

Streamline regulatory requirements across jurisdictions, reducing the compliance burden and legal uncertainties.

✓ **Foster Data Sharing and Interoperability:**

Promote utility of advancements in AI technology by enabling greater data sharing and interoperability.

✓ **Support Learning and Evolution:**

Improve AI reliability and patient outcomes through continual learning and evolution based on real-world data and representative patient populations.

✓ **Address Job Concerns:**

Mitigate job-loss fears by highlighting the value of AI-supported work practices and its healthcare application.



Conclusion

While the barriers described within this report have inhibited progress thus far, removing them is achievable through further public and private sector collaboration. HLC and its members are committed to continue serving as a resource to inform needed AI regulatory and legislative actions.

The risk of inaction is compounding. A patchwork of rules will harden into incompatible and competing systems. Incomplete or biased data will deepen access gaps and degrade model performance. Public trust will erode further adoption. Waiting turns today's friction into tomorrow's failure to scale when safely and effectively scaling AI can better support providers and serve patients.

What is needed now is federal action with urgency and clarity. This report outlines practical, high-impact actions that can be implemented immediately, alongside longer-term objectives that require sustained effort and coordination. If the U.S. can align policy with technology and clinical realities, AI will not only enhance human intelligence—it will extend the reach of better and safer care to all communities.

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Appendix

1. Summary of Interview Methodology

This report is grounded in a series of structured, 30-minute interviews with subject matter experts from HLC member companies that span the healthcare ecosystem. Interviewees were nominated by their organizations to ensure a diversity of perspectives, with representation from regulatory, compliance, data, technology, clinical operations, and digital health innovation functions.

Each interview followed a consistent process focused on exploring barriers and accelerators to AI adoption in healthcare, anchored around three core themes:

- **People:** Trust and adoption of AI, stakeholder awareness and education, and the incentives or disincentives influencing AI uptake.
- **Process & Policy:** Existing standards and regulations, integration challenges, and the operational burden of enabling AI solutions.
- **Data & Technology:** Data interoperability and privacy, data quality and completeness, and the infrastructure required to support AI initiatives.

This approach ensured that the insights were grounded in real-world experience and reflect the collective insights of leading organizations across the healthcare landscape.

2. Participating Organizations

A total of 27 organizations from HLC's membership participated in the interview process, representing a broad cross-section of the healthcare value chain. The organizations interviewed include:

Payers

- The Cigna Group
- GuideWell
- UnitedHealth Group

Providers

- AdventHealth
- City of Hope
- ComPsych
- Envision Healthcare
- Labcorp
- MemorialCare
- Mount Sinai Health System
- Premier
- Sanford Health
- Texas Health Resources
- Vizient
- Wellvana Health

Manufacturers

- Amgen
- Baxter
- Johnson & Johnson
- Novartis
- Pfizer

Data & Technology Partners

- Amazon
- Cotiviti
- eHealth
- Leidos
- Surescripts

Distributors & Accreditors

- Cencora
- Joint Commission

